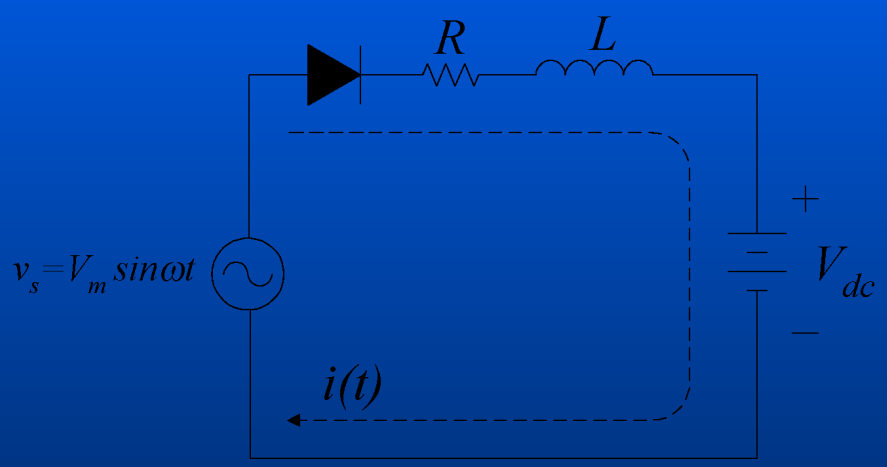
3.5 R-L 기전력부하

$$V_m = 100V$$



$$i = \frac{V_m - V_{dc}}{Z}$$

$$= \frac{100-0}{1} = 100\,A$$

$$i = \frac{100-50}{1} = 50\,A$$

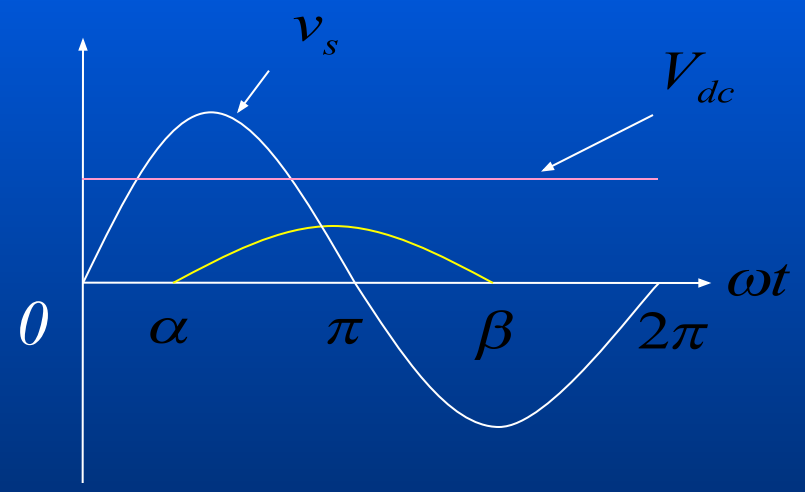
$$i = \frac{100-100}{1} = 0$$

- By KVL

$$V_m \sin\omega t = Ri + L\frac{di}{dt} + V_{dc}$$

1. Battery에 충전전압이 0인 경우 최대전류가 흐름
2. 충전전압이 상승하면 이력 대비 출력 전압이 작아지므로 전류도 감소됨
3. 보호용 BMS(Battery Management System)이 필요함

3.5 R-L 기전력: 전류



$$V_m \sin\alpha = V_{dc}$$

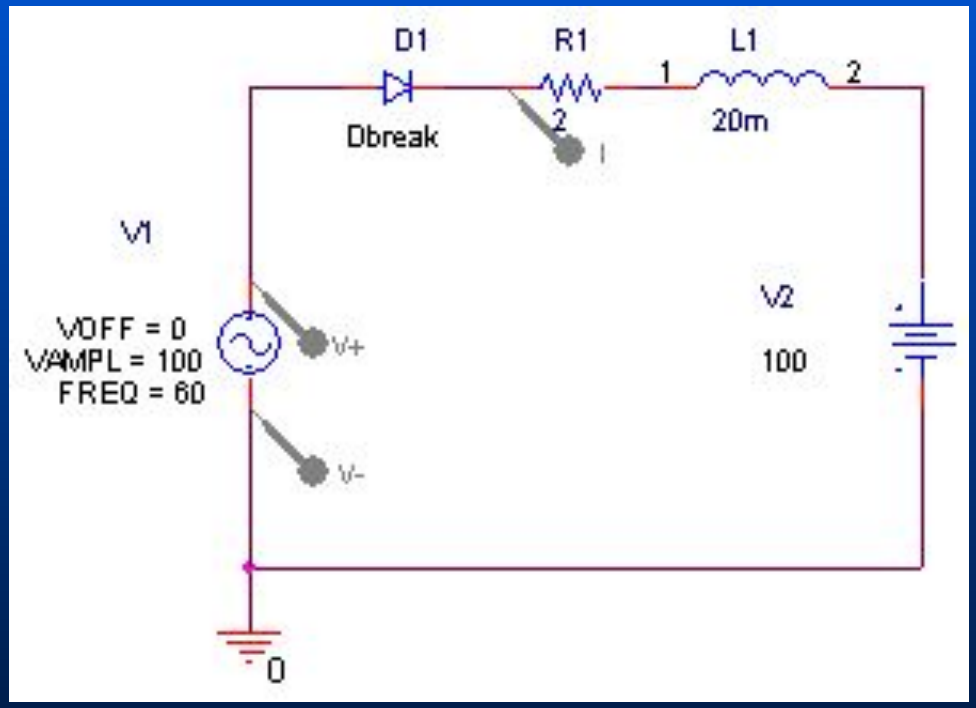
$$\therefore \alpha = \sin^{-1}\left(\frac{V_{dc}}{V_m}\right)$$

- 전류

$$i(\omega t) = \frac{V_m}{Z}\sin(\omega t - \theta) - \frac{V_{dc}}{R} + \left(-\frac{V_m}{Z}\sin(\alpha - \theta) + \frac{V_{dc}}{R}\right)e^{-\frac{\alpha}{\omega\tau}}$$

$$at\ \alpha \le \omega t \le \beta$$

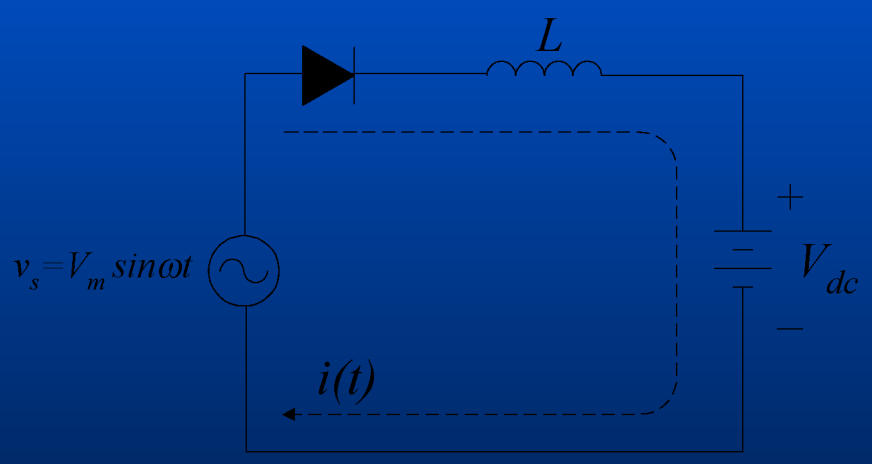
예제 3-5 R-L 기전력부하 반파정류기



- *Parameter*
  - ✔ *R=2 Ω*
  - ✔ *L=20 mH*
  - ✔ *f=ω/2π=60 Hz*
  - ✔ V_m*=169.7 V*
  - ✔ V_{dc}*=100 V*
  - ✔ *Transient:*

    0 0.01m 50ms

3.6 인덕터 기전력부하



- By KVL

$$V_m \sin\omega t = L\frac{di}{dt} + V_{dc}$$

$$V_m \sin\omega t = \omega L\frac{di}{d\omega t} + V_{dc}$$

3.6 인덕터 기전력부하: 전류

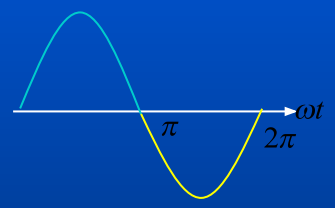
$$V_m \sin\omega t = \omega L \frac{di}{d\omega t} + V_{dc}$$

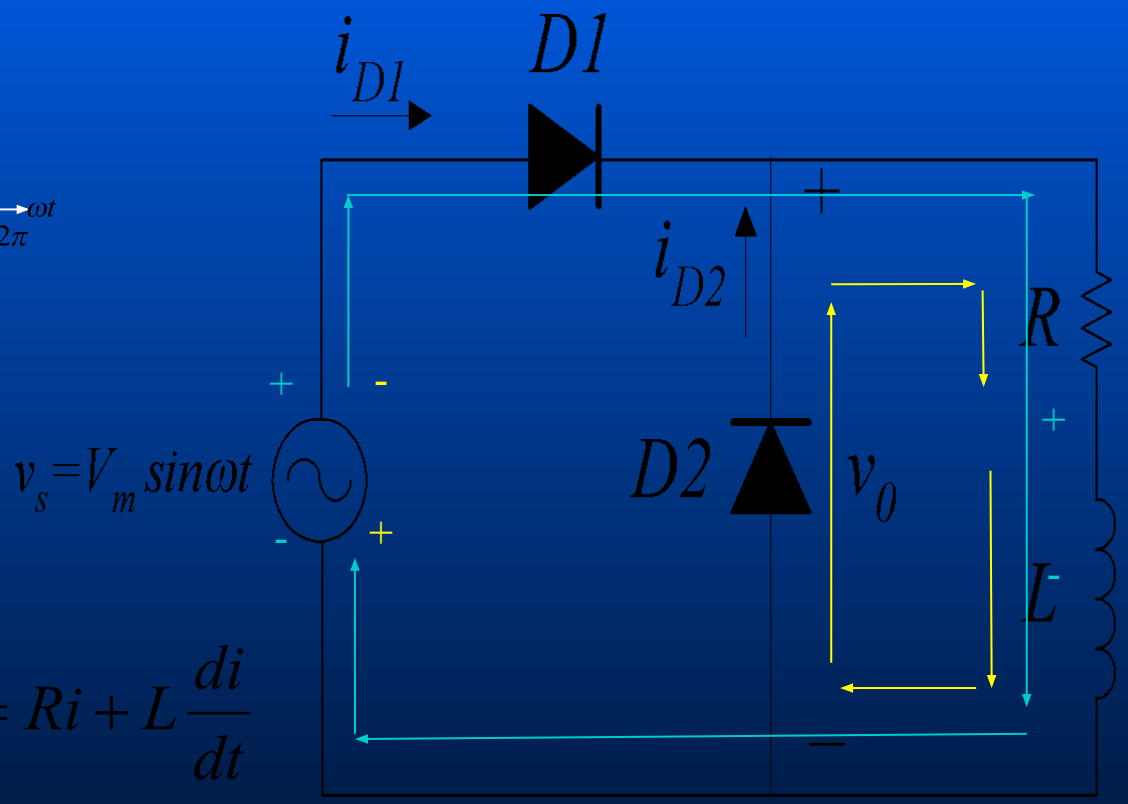
$$\frac{di}{d\omega t} = \frac{V_m \sin\omega t - V_{dc}}{\omega L}$$

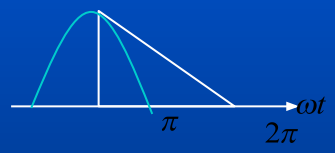
$$i(\omega t) = \frac{1}{\omega L}\int_0^{\omega} V_m \sin\omega t d\omega t - \frac{1}{\omega L}\int_0^{\omega} V_{dc} d\omega t$$

$$i(\omega t) = \begin{cases} \frac{V_m}{\omega L}(\cos\alpha - \cos\omega t) + \frac{V_{dc}}{\omega L}(\alpha - \omega t) & at\ \alpha \le \omega t \le \beta \\ 0 & otherwise \end{cases}$$

3.7 환류 다이오드(Freewheeling Diode)





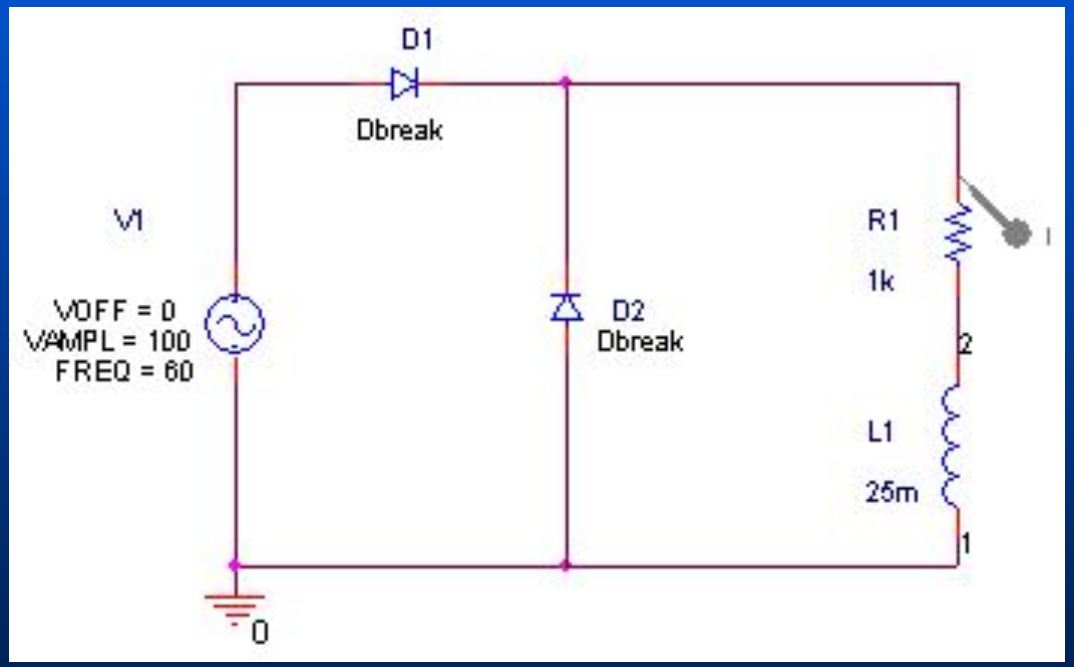


$$V_m \sin\omega t = Ri + L\frac{di}{dt}$$

$$0 = Ri + L\frac{di}{dt}$$

1. +인가시 D1를 통하여 L에 전기가 저장
2. –인가시 L에 저장된 전기가 방전되어 출력을 직류에 가깝도록함

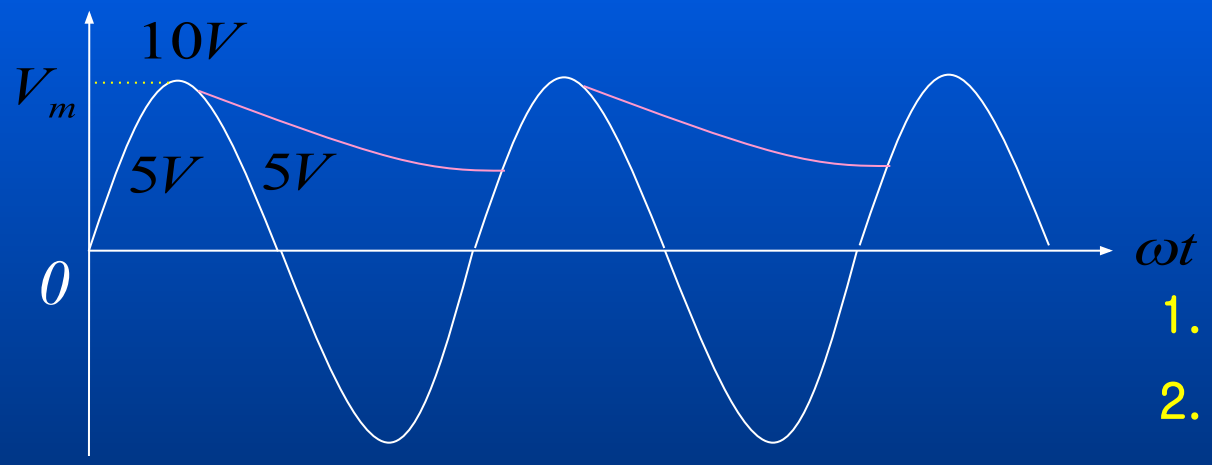
예제 3-7 환류다이오드를 가진 반파정류기

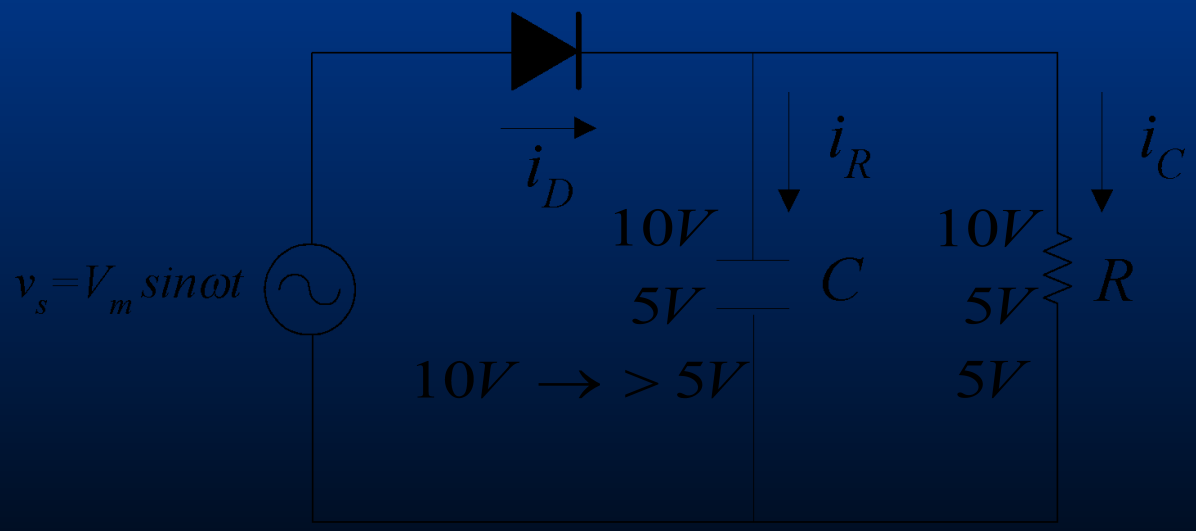


- *Parameter*
  - ✔ *R=2 Ω*
  - ✔ *L=25 mH*
  - ✔ *f=60 Hz*
  - ✔ V_m*=100 V*
  - ✔ *Transient:*

    0 1m 50m

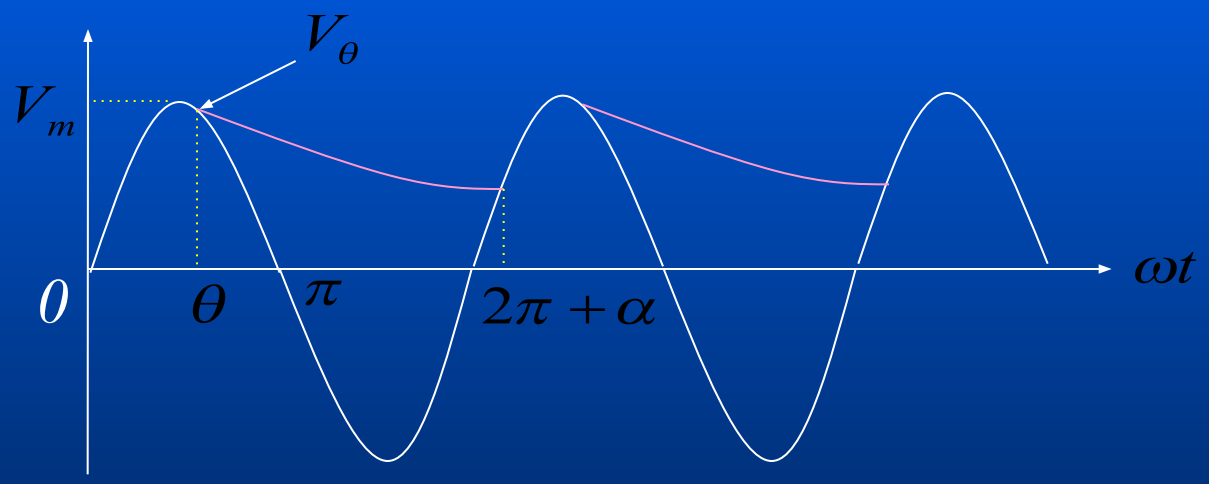
3.8 R-C 반파정류회로





1. 병렬시 전압이 같음
2. 10V 인가시 커패시터와 저항이 전압이 10V로 동일
3. 전압이 10V에서 5V로 될 때 커패시터의 전압이 10V에서 5V가 되어 방전된 전압이 출력을 직류에 가깝도록함

3.8 R-C 반파: 전류(1)



$$V_0(\omega t) = \begin{cases} V_m \sin \omega t & at\ diode\ on \\ V_\theta \cdot e^{-\frac{(\omega t-\theta)}{\omega RC}} & at\ diode\ off \end{cases}$$

*여기서

$$V_\theta = V_m \sin\theta\ ,\quad \theta = \tan^{-1}(R\cdot\omega C) = -\tan^{-1}(R\cdot\omega C)+\pi$$

3.8 R-C 반파: 전류(2)

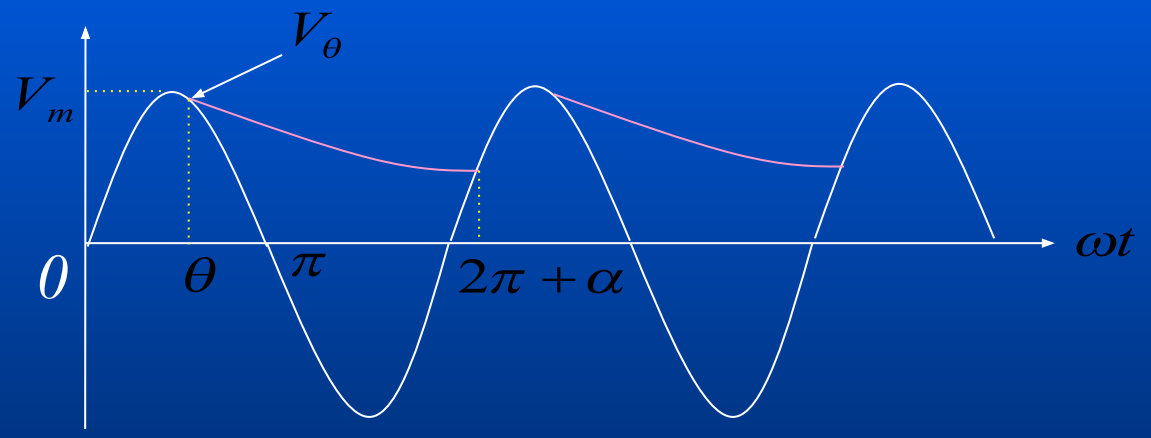
- 점화각 (Firing Angle)

$$for \quad \omega t = 2\pi + \alpha$$

$$V_m \sin(2\pi + \alpha) = (V_m \sin\theta) e^{-\frac{(2\pi + \alpha - \theta)}{\omega RC}}$$

- α의 계산
  - 수치기법(Numerical Analysis) 이용계산
  - OrCAD를 이용하여 계산: 가장 간편함

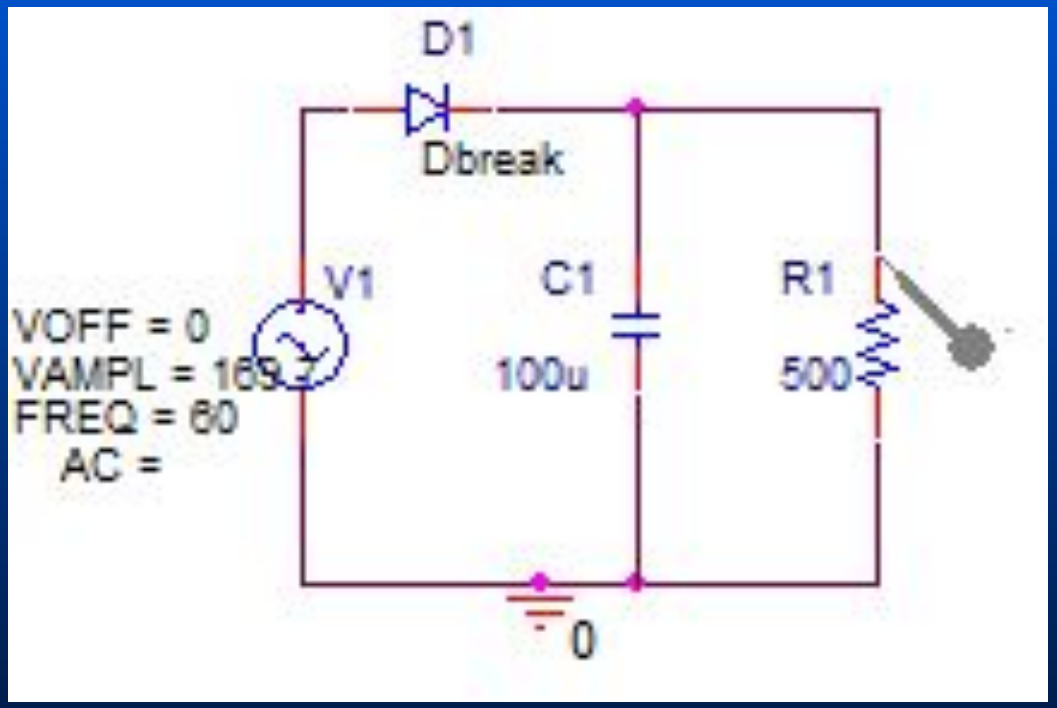
3.8 R-C 반파: 전류(3)



- C의 전류: $i_C = \omega C \dfrac{dV_0(\omega t)}{d(\omega t)}$

$$i_C(\omega t) = \begin{cases} \dfrac{V_m \sin\theta}{R} e^{-\frac{(\omega t-\theta)}{\omega RC}} & at\ 0 \le \omega t \le 2\pi+\alpha \\ \omega C V_m \cos(\omega t) & at\ 2\pi+\alpha \le \omega t \le 2\pi+\theta \end{cases}$$

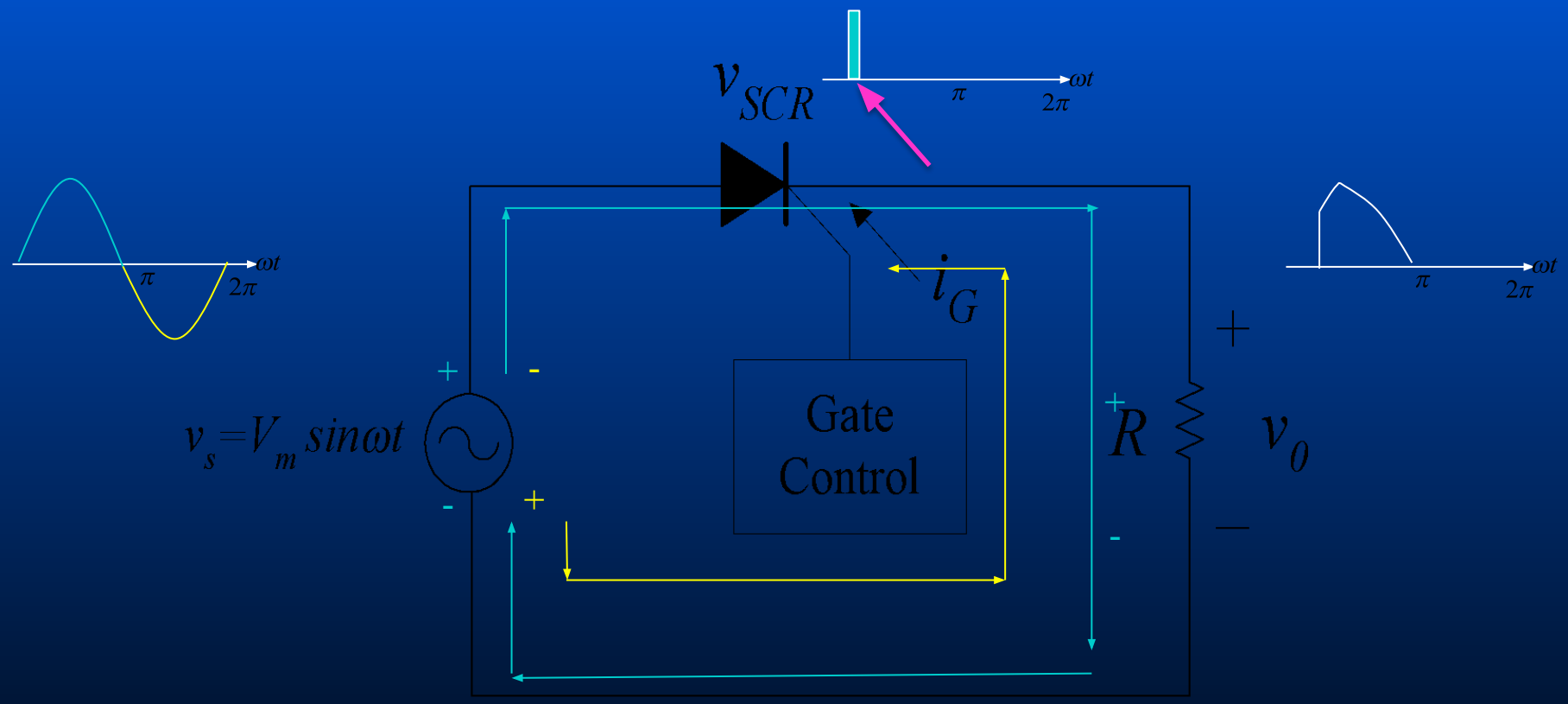
예제 3-9 R-C 부하를 갖는 반파정류기



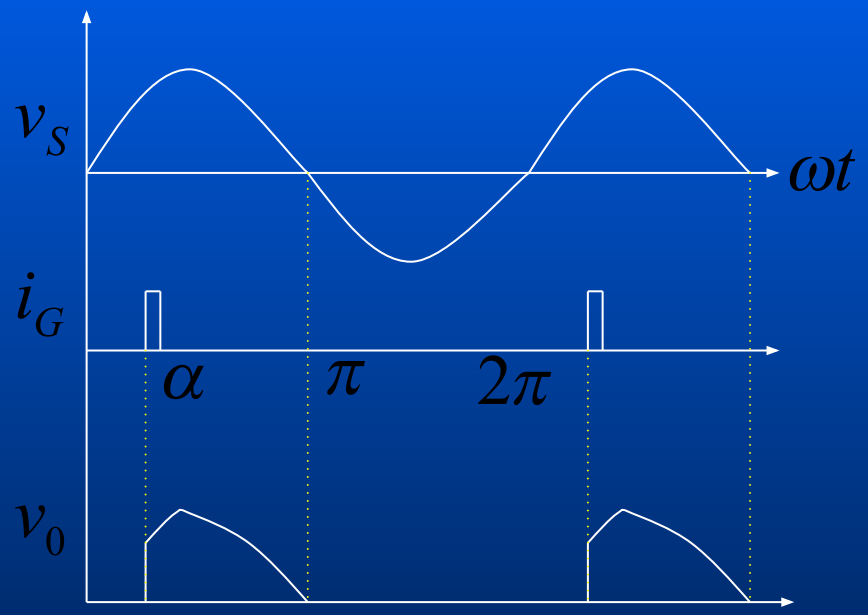
- *Parameter*
  - ✔ *R=500 Ω*
  - ✔ *C=100 μF*
  - ✔ *f=60 Hz*
  - ✔ V_m*=100*
  - ✔ *Transient:*

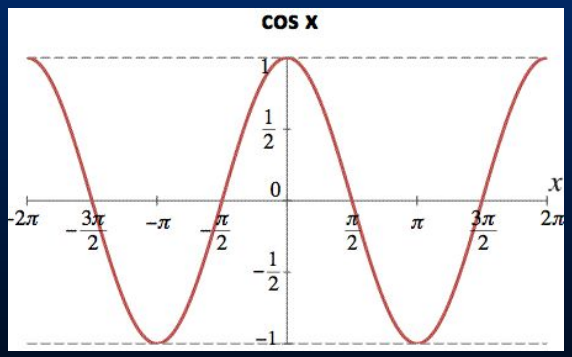
    0 0.01m 50m

3.9 반파제어 정류기



3.9 반파제어 정류기: 저항부하





- 전압의 평균값

$$V_0 = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin\omega t \, d(\omega t)$$

$$\int \sin dt = -\cos t$$

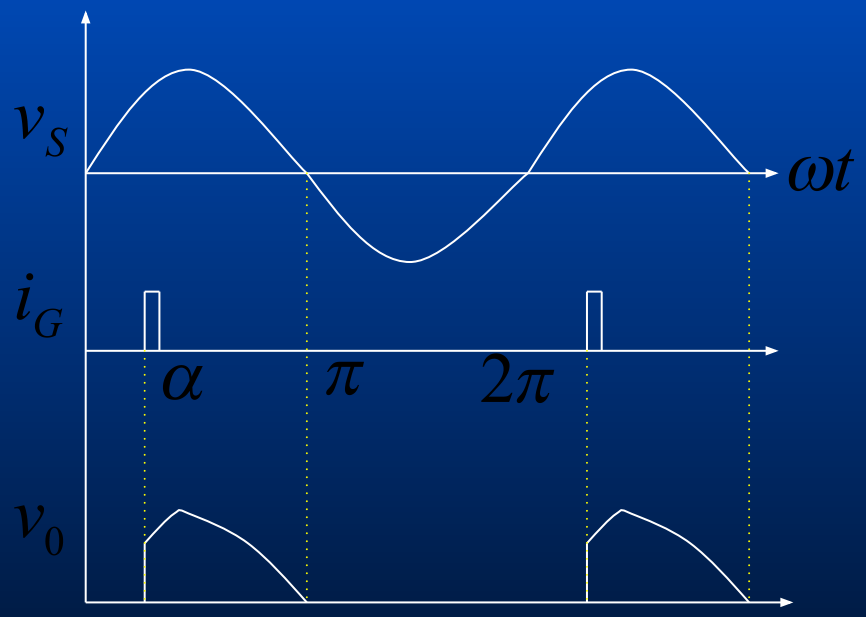
$$= \frac{V_m}{2\pi}\left[-\cos\omega t\right]_{\alpha}^{\pi}$$

$$= \frac{V_m}{2\pi}\left(-\cos\pi + \cos\alpha\right)$$

$$= \frac{V_m}{2\pi}\left(1 + \cos\alpha\right)$$

3.9 반파제어 정류기: 저항부하

- 전압의 실효값



$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi}\left(V_m \sin\omega t\right)^2 d(\omega t)}$$

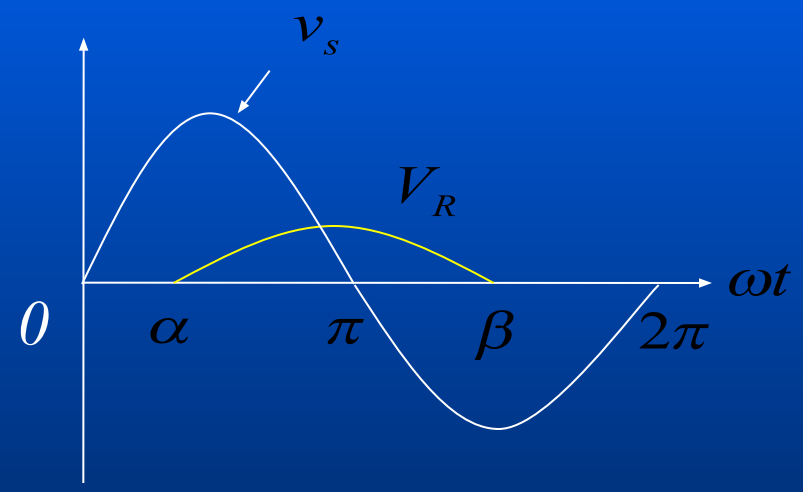
$$\left(\sin^2\omega t = \frac{1-\cos 2\omega t}{2}\right)$$

$$= \sqrt{\frac{V_m^{\ 2}}{2\pi}\left[\frac{\omega t}{2} - \frac{\sin 2\omega t}{4}\right]_{\alpha}^{\pi}}$$

$$= \sqrt{\frac{V_m^{\ 2}}{2\pi}\left(\frac{\pi}{2} - \frac{\sin 2\pi}{4} - \frac{\alpha}{2} + \frac{\sin 2\alpha}{4}\right)}$$

$$= \frac{V_m}{2}\sqrt{\left(1 - \frac{\alpha}{\pi} + \frac{\sin 2\alpha}{2\pi}\right)}$$

3.9 반파제어 정류기: R-L부하

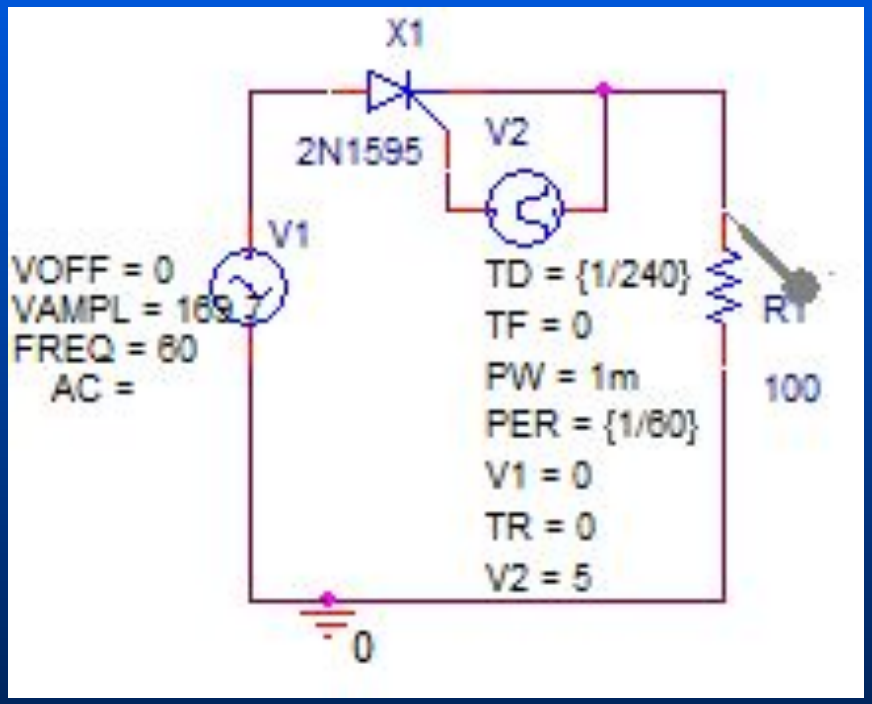


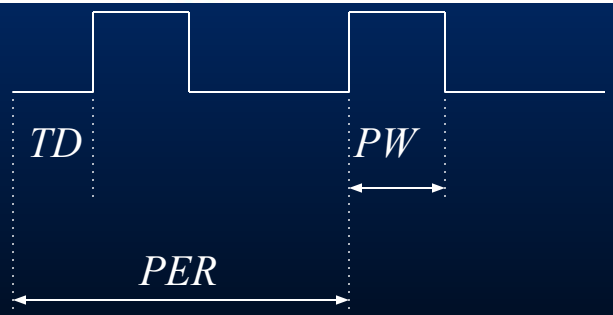
- 전류

$$i(\omega t) = i_f(\omega t) + i_n(\omega t)$$

$$i(\omega t) = \begin{cases} \dfrac{V_m}{Z}\left(\sin(\omega t - \theta) - \sin(\alpha - \theta)e^{-\frac{(\alpha - \omega t)}{\omega \tau}}\right) & at\ \alpha \le \omega t \le \beta \\ 0 & otherwise \end{cases}$$

3.10 OrCAD: 반파제어정류기 설계





- *Parameter*
  - ✔ *R=1 k Ω*
  - ✔ *f=ω/2π=60 Hz*
  - ✔ V_m*=50 V*
  - ✔ V_{pulse}
    - *V1=0 V*
    - *V2=5 V*
    - *TD=0*
    - *PER= 1/60={1/60}*
    - *PW=1m*
  - ✔ *Transient*

    0 0.01m 50m